

Notes: Berg, Burg, Keil & Puri (JFE, 2025)

The economics of “Buy Now, Pay Later”: A merchant’s perspective

Contents

1	Big picture	2
2	Data and measurement	2
2.1	Merchant and product	2
2.2	Randomized experiment	2
2.3	Outcome and cost variables	3
3	Models and empirical specifications	3
3.1	Simple price-discrimination example	3
3.2	Randomized sales regressions	3
3.3	Credit-score heterogeneity	3
3.4	BNPL approval and profitability	4
4	Identification and empirical design	4
5	Tables and figures	4
5.1	Figure 1, Table 1 and Table 2: mix and experimental balance	4
5.2	Figure 2, Table 3, Table 4: BNPL availability, credit score, and sales	5
5.3	Table 5, Figure 3, Table 6: Merchant lending function and costs	5
5.4	Table 7 and Table 8: Overall economics and recession robustness	6
5.5	Figures A.1–A.2 and igure A.3: Appendix model figures	7
6	Short summary	7
7	Conclusion	7
7.1	Core conclusion	7
7.2	Economic intuition	7
7.3	Takeaway	7

1 Big picture

- **Question.** Why do merchants offer zero-interest Buy Now, Pay Later (BNPL) financing, and is it profitable from the merchant’s point of view?
- **Central idea.** BNPL bundles a product with a subsidized short-term loan. Since the subsidy is most valuable to low-creditworthiness customers, BNPL is an implicit targeted price cut.
- **Mechanism.** When creditworthiness is positively correlated with willingness to pay, the merchant can price-discriminate without changing posted prices.
- **Setting.** The authors study a German online furniture merchant that offers an in-house “pay in 30 days” BNPL product.
- **Main experiment.** Randomly making BNPL available raises conversions by about 9 percentage points and raises total revenue per checkout visit by about EUR 41–47, roughly 20 percent.
- **Heterogeneity.** The sales effect is much stronger among low-creditworthiness customers.
- **Lending function.** The merchant approves BNPL more often for customers with higher credit scores and for carts with higher gross margins.
- **Profitability.** BNPL raises costs and defaults, but the added sales dominate. A stylized calculation implies about EUR 3.3 of incremental profit per EUR 100 of cart value.
- **Takeaway.** BNPL is both a payment technology and a pricing technology.

2 Data and measurement

2.1 Merchant and product

- The data come from a German e-commerce company selling furniture through its own website.
- The BNPL product is a zero-interest “pay in 30 days” loan provided in-house by the merchant.
- The merchant observes checkout visits, conversions, payment methods, order values, gross margins, credit scores, and county characteristics.
- The internal credit score runs from 0 to 1, where 0 is highest default risk and 1 is lowest default risk.

Interpretation. The setting is unusually informative because the merchant observes both the sales gains and the credit/default costs of BNPL.

2.2 Randomized experiment

- In early 2022, BNPL availability was randomly shown or not shown at checkout.
- Prices and other payment options were held fixed.
- The treatment group contains roughly 74,000 customers; the unavailable group contains roughly 950 customers.
- The main estimates are reduced-form availability effects, not endogenous usage effects.

Interpretation. The experiment identifies what happens when the merchant introduces BNPL as a payment option.

2.3 Outcome and cost variables

- **Conversion:** a checkout visit becomes a completed purchase.
- **Conditional revenue:** revenue among purchases.
- **Unconditional revenue:** revenue per checkout visit, combining extensive and intensive margins.
- **BNPL approval:** the merchant approves a BNPL application.
- **Default:** BNPL-financed revenue not repaid after reminders and collection transfer.
- **BNPL cost:** capital cost plus default losses; average cost is about 2.1 percent of BNPL revenue.

Interpretation. A merchant-level evaluation must include added sales, default losses, capital costs, and payment-method substitution.

3 Models and empirical specifications

3.1 Simple price-discrimination example

There are two customers, with willingness to pay 100 and 95. The merchant’s marginal cost is 92. Without BNPL,

$$\pi(P = 100) = 1(100 - 92) = 8, \quad \pi(P = 95) = 2(95 - 92) = 6.$$

If the merchant charges 100 but bundles the purchase with a zero-interest BNPL loan, a customer with default probability p faces effective expected payment $100(1 - p)$. If the low-WTP customer has $p = 5\%$, then her effective price is 95 and she buys:

$$\pi_{BNPL}(P = 100) = (100 - 92) + (100(1 - 0.05) - 92) = 8 + 3 = 11.$$

Interpretation. BNPL is an implicit discount targeted toward customers who value credit most.

3.2 Randomized sales regressions

The baseline reduced-form specification is

$$Y_i = \alpha + \beta W_i + \varepsilon_i,$$

where $W_i = 1$ if BNPL is available. The paper also estimates versions with controls and fixed effects:

$$Y_i = \alpha + \beta W_i + \delta X_i + \gamma_c + \theta_d + \sigma_t + \varepsilon_i.$$

Y_i is conversion, conditional revenue, or unconditional revenue.

Interpretation. Because BNPL availability is randomized, controls mainly improve precision and verify balance.

3.3 Credit-score heterogeneity

$$Conversion_i = \alpha + \beta_1 BNPL_i + \beta_2 Score_i + \beta_3 (BNPL_i \times Score_i) + Controls_i + \varepsilon_i.$$

Since higher score means lower default risk, a negative β_3 means BNPL has a larger effect for low-creditworthiness customers.

Interpretation. This tests the core price-discrimination mechanism.

3.4 BNPL approval and profitability

$$Approved_i = \alpha + \theta_1 Score_i + \theta_2 GrossMargin_i + Controls_i + FE + \varepsilon_i.$$

The stylized merchant profit calculation is

$$\Pi_{BNPL} = \Delta R(m - c_{BNPL}) - R_{displaced}(c_{BNPL} - c_{other}).$$

Interpretation. BNPL is profitable when the margin on new revenue exceeds financing/default costs and payment-method displacement costs.

4 Identification and empirical design

- **Randomization.** The strongest causal result comes from randomly varying BNPL availability.
- **Balance.** Treatment and control customers are similar in cart balance, credit score, age, gender, device type, and county characteristics.
- **Intent-to-treat.** The main coefficient is the policy-relevant effect of offering BNPL, not the endogenous effect of choosing BNPL.
- **Heterogeneity.** Stronger effects for low-creditworthiness customers support the price-discrimination interpretation.
- **Approval regressions.** Positive effects of score and margin on approval show that merchant lending decisions reflect expected default costs and product margins.
- **Limits.** The evidence comes from one German merchant and does not fully estimate consumer welfare or behavioral overborrowing.

Interpretation. The paper is most convincing on merchant sales and profitability; consumer welfare requires additional assumptions.

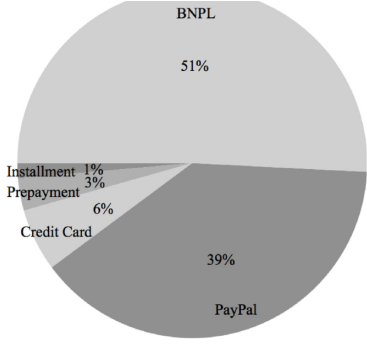
5 Tables and figures

Adjacent figures and tables are grouped to save space and improve reading speed. All visuals are directly cropped from the uploaded paper.

5.1 Figure 1, Table 1 and Table 2: mix and experimental balance

Read: Figure 1 shows BNPL is the dominant payment option when available: 51 percent of payment selections are BNPL, followed by PayPal at 39 percent. Table 1 shows randomized balance. Table 2 shows BNPL users have lower credit scores and look like customers for whom free credit is valuable.

Economic message. BNPL is widely used when offered, and its users match the financing-subsidy mechanism.



g. 1. BNPL and other payment options. This figure plots the shares of Buy-Now-Pay-Later (BNPL) and all other payment options at the e-commerce company. The sample is based on the treatment group in which all payment options are available in the experiment.

(a) Figure 1: payment options

Table 1

Assignment to treatment and control groups.

	Unavailable		Available		p-value
	N (1)	Mean (2)	N (3)	Mean (4)	
Initial Cart Balance (€)	948	392.90	74,124	392.00	(0.953)
Credit score (min. 0, max. 1)	907	0.735	70,155	0.746	(0.122)
Age (Years)	119	43.941	8601	43.445	(0.634)
Male (1/0)	948	0.300	74,124	0.295	(0.759)
Returning customer (1/0)	948	0.308	74,124	0.315	(0.663)
System: Apple (1/0)	948	0.423	74,124	0.447	(0.141)
County Mean Income (€)	940	22,028	73,696	22,192*	(0.068)
County Population Density	940	1152	73,696	1212	(0.171)

This table compares selected characteristics of customers randomly assigned to the group in which BNPL was either unavailable (Columns 1-2) or available (3-5). See Appendix Table A.1 for details on the definition of variables and A.2 for a comparison of all variables and more descriptive statistics. ***, **, and * indicate statistical significance at 1%, 5%, and 10%.

(b) Table 1: randomization balance

Table 2

Customers Selecting BNPL and Other customers.

	Other customers		Selecting BNPL		p-value
	N (1)	Mean (2)	N (3)	Mean (4)	
Initial Cart Balance (€)	20,443	352.06	23,348	375.84***	(0.000)
Credit Score (min. 0, max. 1)	19,195	0.710	22,982	0.706*	(0.056)
Age (Years)	178	42.191	222	40.248*	(0.086)
Male (1/0)	20,443	0.346	23,348	0.232***	(0.000)
System: Apple (1/0)	20,443	0.436	23,348	0.405***	(0.000)
County Mean Income (€)	20,311	22,234	23,215	22,120***	(0.000)
County Population Density	20,311	1309	23,215	1039***	(0.000)

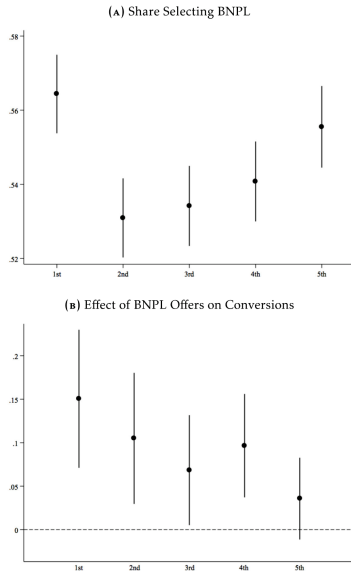
This table compares selected characteristics of customers reaching the check-out website and selecting either PayPal, credit card, pre-payment, or installments (Columns 1-2) or BNPL (3-5). The sample is based on the treatment group (for which all payment options are available) in the experiment. See Appendix Table A.1 for details on the definition of variables. ***, **, and * indicate statistical significance at 1%, 5%, and 10%. P-values are between parentheses. Note that the sample is based on first-time customers, as information on payment method selections for customers that visited the website before is unreliable.

(c) Table 2: customer selection

5.2 Figure 2, Table 3, Table 4: BNPL availability, credit score, and sales

Read: Figure 2 shows the BNPL conversion effect declines with credit score. Table 3 shows BNPL availability raises conversion by about 8.6–9.1 percentage points and unconditional revenue by about EUR 40.8–47.2. Table 4 shows the interaction between BNPL and credit score is negative and significant.

Economic message. The randomized sales effect is large and concentrated where the financing subsidy should matter most.



(a) Figure 2: credit score heterogeneity

Table 3

Effects of BNPL offers on sales.

Panel (A) Dependent variable: Conversion (1/0)			
	(1)	(2)	(3)
BNPL offered (1/0)	0.091*** (0.015)	0.086*** (0.014)	0.086*** (0.014)
Observations	75,072	75,072	75,072
Panel (B) Dependent variable: €-Revenue Conditional on a Conversion			
BNPL offered (1/0)	24.530* (12.942)	15.811** (6.345)	13.720** (6.357)
Observations	60,036	60,036	60,036
Panel (C) Dependent variable: €-Revenue Unconditional on a Conversion			
BNPL offered (1/0)	47.156*** (10.286)	44.963*** (9.008)	40.806*** (9.123)
Observations	75,072	75,072	75,072
Controls			
Customer		Yes	Yes
County		Yes	–
Fixed Effects			
County			Yes
Date			Yes
Time-of-Day			Yes

Regressions in Panel A explain the likelihood that a customer's shopping visit at the e-commerce company's website is converted into a purchase (the extensive margin). Results are linear probability estimates. In Panel B (C) the dependent variable is total realized revenue by customer conditional (unconditional) of a conversion taking place as a measure of the intensive (total) margin. The independent variable of interest is the treatment dummy equal to 1 (0) if BNPL was made available (unavailable) for a randomly selected customer. See Appendix Table A.1 for details on the definition of variables. ***, **, and * indicate statistical significance at 1%, 5%, and 10% for robust White standard errors (between parentheses). "–" indicates that controls are absorbed by fixed effects.

(b) Table 3: BNPL effects on sales

Table 4

Differential effect of BNPL on conversions by credit score.

Dependent Variable: Conversion (1/0)			
	(1)	(2)	(3)
BNPL offered × Credit score	–0.153** (0.072)	–0.154** (0.069)	–0.143** (0.069)
BNPL offered (1/0)	0.198*** (0.057)	0.197*** (0.055)	0.190*** (0.054)
Credit score (min. 0, max. 1)	0.644*** (0.071)	0.643*** (0.069)	0.611*** (0.069)
Observations	71,061	71,061	71,061
Controls			
Customer		Yes	Yes
County		Yes	–
Fixed Effects			
County			Yes
Date			Yes
Time-of-Day			Yes

Regressions explain the likelihood that a customer's shopping visit at the e-commerce company's website is converted into a purchase. Results are linear probability estimates. The independent variables of interest are the treatment dummy equal to 1 (0) if BNPL was made available (unavailable) for a randomly selected customer, the internal credit score (ranging from 0 to 1, where larger values correspond to lower default likelihoods), and their interaction term. See Appendix Table A.1 for details on the definition of variables. ***, **, and * indicate statistical significance at 1%, 5%, and 10% for robust White standard errors (between parentheses). "–" indicates that controls are absorbed by fixed effects.

(c) Table 4: heterogeneity by credit score

5.3 Table 5, Figure 3, Table 6: Merchant lending function and costs

Read: Table 5 shows approvals rise with credit score and gross margin. Figure 3 shows default rates decline steeply with creditworthiness. Table 6 shows BNPL raises variable payment costs because it creates new BNPL revenue and displaces cheaper payment methods.

Economic message. The merchant acts like a lender: it screens borrowers and cares about margins.

Table 5

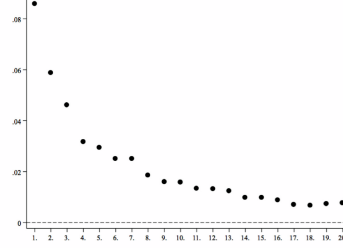
Effects of profit margins and credit scores on BNPL approvals.

Dependent variable: BNPL application approved (1/0)

	(1)	(2)	(3)	(4)
Score (min. 0, max. 1)	0.679*** (0.039)	0.629*** (0.038)	0.651*** (0.037)	0.658*** (0.036)
Gross Margin (in %)	0.065*** (0.020)	0.084*** (0.020)	0.081*** (0.018)	0.093*** (0.009)
Observations	590,994	590,994	590,994	224,570
Controls				
Customer		Yes	Yes	Yes
County		Yes	–	–
Fixed Effects				
County			Yes	Yes
Date			Yes	Yes
Time-of-Day			Yes	Yes
Cart Item Categories			Yes	Yes
Customer			Yes	Yes

Regressions explain the likelihood that the e-commerce company approves a customer's application for using BNPL. Results are linear probability estimates. The independent variables of interest are the customer's credit score (ranging from a high default risk value of 0 to a low default risk value of 1) and the gross profit margin (revenue – purchase costs) of customer's shopping cart at time t . The estimation sample includes all customers applying for BNPL in 2021. See Appendix Table A.1 for details on the definition of variables. ***, **, and * indicate statistical significance at 1%, 5%, and 10% for standard errors (between parentheses) that allow for clustering by product category and county. "–" indicates that controls are absorbed by fixed effects.

(a) Table 5: approvals



(b) Figure 3: defaults

Table 6
Displacement costs.

Option	Costs in % (1)	Revenue per € 100		Costs × Revenue per € 100	
		Unavailable (2)	Available (3)	Unavailable (4)	Available (5)
BNPL					
New Revenue	2.1	0.0	21	0.00	0.44
Displaced Revenue	2.1	0.0	38.9	0.00	0.82
Other	0.9	100	61.3	0.93	0.65
PayPal	1.0	67.9	43.9	0.68	0.44
Credit Card	1.3	13.3	8.2	0.17	0.11
Installment	2.8	2.7	3.7	0.08	0.10
Prepayment	0.0	16.0	5.4	0.00	0.00
Sum (BNPL Displaced + Other)		100	100	0.93	1.47
Sum (BNPL All + Other)		100	121	0.93	1.91

Column (1) contains variable transaction costs, expressed in % of transaction volume. Information comes from both external sources and the merchant. BNPL costs include losses due to unrecovered defaults and capital costs. Payment option use when BNPL is either not shown or shown is in Columns (2) and (3), respectively and expressed per € 100 of purchase volume. Column (4) is the product of Columns (1) and (2), representing total variable transaction costs per € 100 purchase volume when BNPL is unavailable. Column (5) is the same number for when BNPL is available. BNPL - New Revenue in Column (5) is the estimate from the effect on the total margin in Table 5, scaled to € 100. BNPL - Displaced Revenue in Column (5) is the share of BNPL when this option is available, minus the new revenue generated by BNPL.

(c) Table 6: displacement costs

5.4 Table 7 and Table 8: Overall economics and recession robustness

Read: Table 7 implies mean incremental profit of about EUR 3.3 per EUR 100 of cart value. Profitability is strongest in middle credit-score groups and weakest for the riskiest 1 percent. Table 8 shows recession counties have higher BNPL defaults, but the increase is small relative to profits.

Economic message. BNPL is profitable on average, but screening and product margins are crucial.

Table 7

Economics of BNPL.

Per € 100	Mean	Quintiles					p1
		Q5 (2)	Q4 (3)	Q3 (4)	Q2 (5)	Q1 (6)	
ΔR	€ 21	€ 14	€ 17	€ 20	€ 24	€ 30	€ 35
m	20%	20%	20%	20%	20%	20%	20%
c_{BNPL}	2.1%	1.0%	1.2%	1.4%	1.9%	4.3%	8.2%
π on new R	€ 3.8	€ 2.7	€ 3.2	€ 3.7	€ 4.3	€ 4.7	€ 4.1
$R_{displaced}$	€ 39	€ 39	€ 39	€ 39	€ 39	€ 39	€ 39
c_{BNPL}	2.1%	1.1%	1.1%	1.3%	1.7%	3.9%	8.2%
c_{other}	0.9%	0.9%	0.9%	0.9%	0.9%	0.9%	0.9%
$\Delta \pi$ on old R	€ 0.5	€ 0.1	€ 0.1	€ 0.2	€ 0.3	€ 1.2	€ 2.9
Π_{BNPL}	€ 3.3	€ 2.6	€ 3.1	€ 3.5	€ 3.9	€ 3.4	€ 1.3

This table illustrates the overall economics of offering BNPL in a simplified and stylized way. Rows contain values for the respective variable in equation $\Pi_{BNPL} = \Delta R \cdot (m - c_{BNPL}) - R_{displaced} \cdot (c_{BNPL} - c_{other})$. $\Delta R \cdot (m - c_{BNPL})$ is the profit on new revenue generated by BNPL. $R_{displaced} \cdot (c_{BNPL} - c_{other})$ is the change in profit on revenue that would have occurred without BNPL, where customers chose to use the more expensive BNPL if available (without altering their shopping decisions). Column 1 contains mean values for the entire sample. Means by bin for credit score quintiles are in Columns 2-6. Q5 is the highest quintile with the best credit scores. The worst 1% are in Column 7. All €-values are expressed on the basis of € 100 of shopping cart value at check out. For simplicity we assume constant margins, periods between order and payment, displacement revenue, costs of alternative payment methods. We instead focus on the dominating drivers behind the credit score heterogeneity: differential effects on revenue and differential losses due to payment default. The value in Column 1 for ΔR comes from Panel C, Table 3. For the other Columns it is based on calculations not included in any table or figure (available upon request). The value in Column 1 for c_{BNPL} comes from Table 6 (additional calculations for the other Columns; available upon request). We assume a value for m for a standard product. The value for $R_{displaced}$ comes from Table 6. c_{other} it comes from Table 6.

Table 8

(a) Table 7: economics of BNPL

Table 8
Dependent variable: BNPL defaults (1/0)

Panel (A) Counties with negative GDP growth				
	(1)	(2)	(3)	(4)
Recession County (1/0)	0.003*** (0.001)	0.002** (0.001)	0.001 (0.001)	0.001 (0.001)
Observations	1588	1588	1588	1588
Recession Counties	106	106	106	106
Panel (B) Counties with GDP Growth < -2%				
Recession County (1/0)	0.003* (0.002)	0.002 (0.002)	0.003* (0.002)	0.002 (0.002)
Observations	1588	1588	1588	1588
Recession Counties	28	28	28	28
Fixed Effects				
Year Fixed Effects		Yes		Yes
County Fixed Effects			Yes	Yes

Regressions in this table are estimated from a county-year panel with 400 counties between 2016 and 2019. The independent dummy variable identifies recession counties. It equals 1 if annual GDP growth is negative. The dependent variable in Panel (A) is the share of BNPL financed conversions that end up in default. In Panel (B) it is the share of customers selecting BNPL (in % of all customers selecting some payment option). ***, **, and * indicate statistical significance at 1%, 5%, and 10% for robust White standard errors (between parentheses). "–" indicates that controls are absorbed by fixed effects.

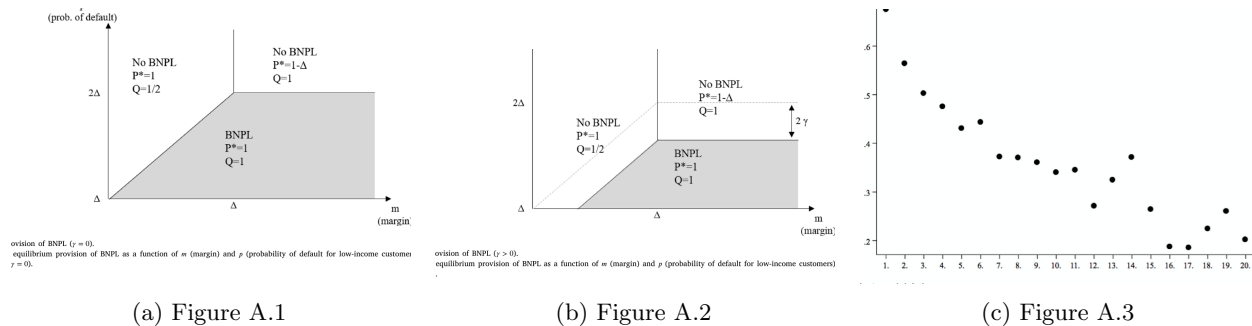
Table 8
Combined control on recession counties: recession counties have higher BNPL defaults

(b) Table 8: recession defaults

5.5 Figures A.1–A.2 and figure A.3: Appendix model figures

Read: Figures A.1–A.2 illustrate BNPL provision regions as a function of product margin and default probability. Figure A.3 shows unrecovered sales in defaults also fall with creditworthiness.

Economic message. BNPL requires enough margin to cover credit losses and operational costs.



6 Short summary

- BNPL is merchant-subsidized short-term credit bundled with a product sale.
- The bundle gives an implicit discount to customers with high borrowing costs or high default risk.
- The randomized experiment shows that BNPL availability increases conversions and sales.
- The effect is strongest among low-creditworthiness customers.
- The merchant approves BNPL more often for high-score customers and high-margin carts.
- BNPL raises payment costs and default risk, but added sales dominate in the merchant's profit calculation.
- Profitability is inverse-U-shaped across creditworthiness.
- External validity is strongest when BNPL financing is paid for mostly by merchants rather than customers.

7 Conclusion

7.1 Core conclusion

BNPL can be understood as a merchant price-discrimination technology: it lowers the effective price for customers who value free credit, while preserving the posted product price.

7.2 Economic intuition

The merchant wants to reach low-WTP customers without cutting prices for everyone. BNPL accomplishes this by embedding the discount in a zero-interest loan. The strategy works when the added margin on new sales exceeds default, capital, and payment-method substitution costs.

7.3 Takeaway

BNPL is not just consumer finance. It is a pricing, payments, and lending technology. For merchants, it can increase profits; for policymakers, it raises questions about distribution and overborrowing.